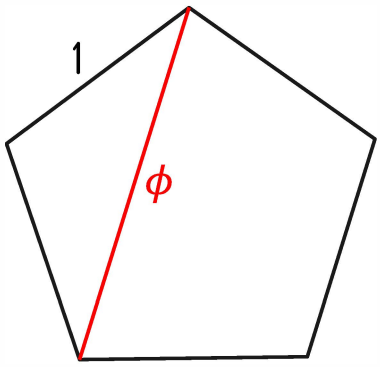
MathsJam Shout

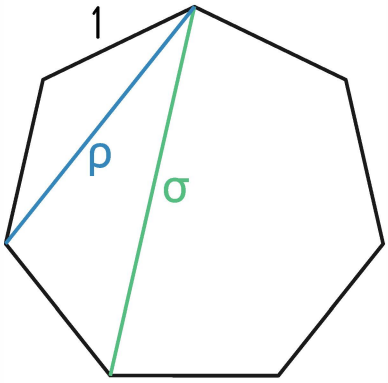
September 2026
Pavia MathsJam, Italy

Puzzle
Diagonals



In a unit pentagon, the diagonal is the golden ratio ϕ, and it satisfies $\phi^2 = \phi + 1$. Since the segments in the "fan" joining each vertex to the other four have lengths equal to 1, ϕ, ϕ, 1, this means that the product of two diagonals equals the sum of every other term in the fan.

Can you work out similar formulas for the heptagon? And can you generalize this for every odd n-gon?

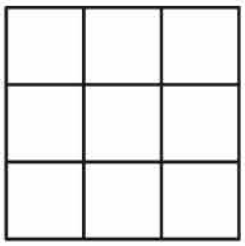


Play
Lanrick

This is a short board game created by Lewis Carroll, of Alice in Wonderland fame. The game is played on a chessboard with only five (later 8) tokens per player and is reminiscent of the elimination mechanism in musical chairs.

Rules (PDF): mathsj.am/lanrick

Puzzle
"Less-than-magic" square



Place nine numbers in arithmetic progression in a 3x3 grid so that the sum of a corner number and its two orthogonal neighbours is constant. You may want to start with the first nine positive integers, but try to generalize this to any arithmetic progression!
(From Italian mathematician Ghersi, 1913)

Puzzle
Inverse sums

Find all quadruples of non-zero real numbers satisfying

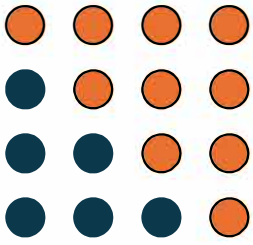
$$a + b + c = d$$

and $$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{1}{d}$$

(From the 1977 Putnam Competition)

Explore
Triangulation

Triangular numbers are numbers of dots that can be arranged in a triangular pattern. Likewise, polygonal numbers count dots arranged in successive regular polygonal patterns. A square number can be decomposed as the sum of two triangular numbers, as shown.



Does a similar relationship hold for other polygonal numbers?

In other words, can every polygonal number be expressed as a sum of triangular numbers?

More info: mathsj.am/polygonal-triangular

MathsJam Shout is a monthly sheet of ideas for activities to do at a MathsJam night. It's created using suggestions from a different MathsJam each month, and if you'd like to submit suggestions for a month in the future, email **katie@mathsjam.com**

Maths Jam is a monthly opportunity for like-minded self-confessed maths enthusiasts to get together in a pub and share stuff they like. Puzzles, games, problems, or just anything they think is cool or interesting. Monthly Maths Jam nights happen in over 70 locations around the world, on the second-to-last Tuesday of each month. To find your nearest Maths Jam, visit the website at **mathsjam.com.**